SinkAware: Approximate Low-Rank Fixed-Sink Priors for Attention

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Abstract

Attention sinks are fixed early tokens that often receive substantial attention. A tempting systems idea is to precompute or reuse their contribution. We show that exact static reuse is invalid under standard softmax attention because fixed sink-token logits remain query-dependent. We therefore test a narrower approximation: per-head low-rank predictors for fixed sink logits while keeping all non-sink scores exact. On Mac-local distilgpt2 traces, rank-2 prediction reduces aggregate held-out attention-output relative L2 error to 0.141 versus 0.170 for a position-only predictor, while the cost model estimates 0.531x exact four-sink QK multiply-adds. Layer-head and head-selection analyses weaken broad claims, but split/seed, length, trace-level, and GPT2/OPT-family gates keep the all-head rank-2 row positive. The strongest Mac-local stability gate keeps all four model/length rows positive at lengths 64 and 96, with mean output relative-L2 improvement 0.0535 ± 0.0262 and minimum row improvement 0.0301. The approximate operator passes Triton interpreter correctness tests against its CPU reference. Larger downstream causal-LM control gates on 48 traces and three split seeds now pass for sink counts 2 and 4 at lengths 64 and 96: exact replacement is a no-op, and rank-2 is closer than position-only in loss drift and KL on both distilgpt2 and OPT-125M. At length 96 with two sink tokens, aggregate absolute loss-delta improvement is 0.0537 ± 0.0509; with four sink tokens it is 0.0721 ± 0.0676. Top-1 disagreement remains non-negligible, so this is not cross-model predictor transfer, benchmark success, or a GPU speed result. It is a bounded correctness, stability, and downstream-control gate for deciding whether approximate sink prediction is worth native implementation.

1 Problem

For query q_t, sink keys K_s, and non-sink keys K_n, the sink logits $q_t K_s^\top$ are still query-dependent. Any exact method that skips them changes the softmax denominator or attention weights. SinkAware kills that exact static-prior branch and revives only an approximate one.

2 Approximate Branch

The live branch predicts fixed sink-token logits from a low-dimensional query feature. Non-sink logits remain exact. The branch is useful only if the approximation is cheap enough and attention-output drift stays bounded. Table 1 reports the original aggregate layer means. This view keeps rank-2 alive: it improves output relative L2 and sink-mass error over position-only while staying under the simple multiply-add estimate for exact four-sink QK.

Table 2 is the main caveat. The same probe now reports layer-head paired improvements over the position-only predictor. Positive values mean lower error than position-only. Rank-2's mean output improvement is positive, but the interval overlaps zero and the win rate is only 0.278. This makes the branch weakly alive, not established.

Predictor	Sink RMSE	Sink-mass MAE	Attention L1	Output rel-L2
static	1229.509	0.095	0.217	0.193
position	1197.390	0.076	0.175	0.170
rank1	1079.541	0.067	0.157	0.160
rank2	1001.506	0.055	0.135	0.141
rank4	838.320	0.045	0.115	0.122
rank8	699.232	0.040	0.104	0.107

Table 1: 48-trace distilgpt2 softmax/output probe. Lower is better.

Predictor	Output rel-L2 improve	95% CI	Win rate	Sink-mass improve
rank1	0.0257	0.0290	0.250	0.0086
rank2	0.0297	0.0378	0.278	0.0206
rank4	0.0596	0.0465	0.333	0.0308
rank8	0.0955	0.0805	0.347	0.0360

Table 2: Layer-head paired improvements over position-only across 72 layer-head cells. Positive is better; intervals are normal 95% intervals over layer-head cells.

We also tested the obvious Mac-local rescue: select rank-2 only for heads where rank-2 beats position-only on a validation split, then evaluate that mixed policy on held-out tokens. The rule selected 19/72 heads but failed held-out: output relative L2 was 0.204 for validation-selected rank-2, compared with 0.172 for position-only and 0.142 for all-rank2. This rules out simple validation head selection as a pre-GPU rescue.

The next stability gate repeated all-head rank-2 under randomized token train/test splits. Across seeds 0, 1, and 2, rank-2 beat position-only on every split, with mean output relative-L2 improvement 0.0368 ± 0.0006, sink-mass MAE improvement 0.0203 ± 0.0003, and attention-L1 improvement 0.0435 ± 0.0004. This supports the all-head aggregate row enough for interpreter work, but not a per-head robustness claim: the output win rate across layer-head cells remains 0.282 ± 0.024.

We then ran a bounded sequence-length and sink-count sweep. For max lengths 64 and 96, sink counts 2 and 4, and three token split seeds per configuration, all configurations kept rank-2 positive against position-only. Mean output relative-L2 improvement across configurations was 0.0366 ± 0.0024, and the minimum configuration improvement was 0.0342. This makes the all-head row more repeatable, but the per-head caveat persists: layer-head output win rate averaged only 0.286 ± 0.010.

Finally, we ran a larger trace-level frozen split repeat. Across three deterministic splits of 48 traces, each split trained on 32 whole traces and evaluated on 16 held-out traces. Rank-2 beat position-only on every split: output relative-L2 improvement averaged 0.0379 ± 0.0014, sink-mass MAE improvement averaged 0.0220 ± 0.0008, attention-L1 improvement averaged 0.0461 ± 0.0014, and the minimum output improvement was 0.0367. This is a stronger repeatability check than token-level splitting, but it does not remove the per-head caveat because the layer-head output win rate remains only 0.278 ± 0.016.

Before local Triton execution was unblocked, we ran a larger held-out/model-family falsification repeat. The gate fits all-head rank-2 predictors separately per model, then compares rank-2 against position-only on held-out traces. On 48 traces and split seeds 0, 1, and 2, distilgpt2 improves by a measured 0.0306 ± 0.0023 output relative-L2, with minimum split improvement 0.0283. facebook/opt-125m improves by a measured 0.0788 ± 0.0069, with minimum split improvement 0.0731. The aggregate model-row improvement is 0.0547 ± 0.0472, with minimum model improvement 0.0306. This is not cross-model predictor transfer and is not promotion evidence, but it is a stronger falsification check than the prior one-seed smoke because the rank-2 row did not collapse under repeated OPT-family held-out splits.

We then broadened that same family gate across sequence length. With 48 traces, lengths 64 and 96, sink count 4, and split seeds 0, 1, and 2, every model/length row remained positive. The aggregate model/length output relative-L2 improvement was 0.0535 ± 0.0262, with minimum row improvement 0.0301 and layer-head output win rate 0.982 ± 0.008. The per-row improvements were 0.0306 ± 0.0023 and 0.0301 ± 0.0018 for distilgpt2 at lengths 64 and 96, and 0.0788 ± 0.0069 and 0.0744 ± 0.0040 for facebook/opt-125m at lengths 64 and 96. This strengthens the Mac-local stability decision surface, but it still does not measure downstream quality, predictor transfer, or GPU speed.

Downstream control gate. We also patch full GPT2/OPT attention during causal-LM evaluation. It compares exact baseline attention against exact sink-logit replacement, position-only replacement, and rank-2 replacement. Exact replacement is the validity control and reproduces baseline loss and logits on both tested models. The latest 48-trace repeats keep rank-2 closer than position-only on both models, both lengths, and sink counts 2 and 4. At length 64 with two sink tokens, aggregate absolute loss-delta improvement is 0.0621 ± 0.0701 and aggregate KL improvement is 0.0532 ± 0.0772, with minimum model loss improvement 0.0263. At length 96 with two sink tokens, the corresponding numbers are 0.0537 ± 0.0509, 0.0464 ± 0.0580, and 0.0277. Four-sink repeats are larger: length 64 gives 0.0801 ± 0.0894 loss improvement and 0.0752 ± 0.1003 KL improvement, with minimum model loss improvement 0.0345; length 96 gives 0.0721 ± 0.0676 and 0.0694 ± 0.0823, with minimum model loss improvement 0.0376. This keeps the branch alive beyond isolated attention-output drift, but only as a downstream-control diagnostic: predictors are still fit separately per model/config/split, top-1 disagreement remains around 0.07–0.15 for rank-2 in the larger rows, and no benchmark accuracy or GPU speed is measured.

Max len	Sink tokens	Loss improve	KL improve	Min model improve
64	2	0.0544 ± 0.0505	0.0485 ± 0.0693	0.0287
64	4	0.0809 ± 0.0815	0.0825 ± 0.1078	0.0393
96	2	0.0433 ± 0.0317	0.0408 ± 0.0541	0.0272
96	4	0.0728 ± 0.0619	0.0769 ± 0.0904	0.0412

Table 3: Downstream causal-LM control sweep on 24 traces, seeds 0/1/2, and separately fit distilgpt2/OPT-125M predictors. Positive values mean rank-2 is closer to exact baseline than position-only.

Traces	Max len	Sink tokens	Loss improve	KL improve	Min model improve
48	64	4	0.0801 ± 0.0894	0.0752 ± 0.1003	0.0345
48	96	4	0.0721 ± 0.0676	0.0694 ± 0.0823	0.0376
48	64	2	0.0621 ± 0.0701	0.0532 ± 0.0772	0.0263
48	96	2	0.0537 ± 0.0509	0.0464 ± 0.0580	0.0277

Table 4: Larger downstream causal-LM control repeats. Exact sink-logit replacement is a no-op in all rows; rank-2 beats position-only in loss drift and KL for both distilgpt2 and OPT-125M across seeds 0/1/2.

As a final Mac-side frontier check, we reran the downstream patch gate on 48 traces at length 96 and sink count 4 with ranks 1, 2, 4, and 8. Higher ranks monotonically reduce downstream drift: aggregate absolute loss delta falls from 0.137 at rank 1 to 0.096 at rank 2, 0.062 at rank 4, and 0.044 at rank 8, while top-1 disagreement falls from 0.143 to 0.125, 0.095, and 0.080. The systems cost model keeps rank 2 as the live compromise because rank 4 and rank 8 are estimated at 1.062x and 2.125x exact four-sink QK multiply-adds. Thus the frontier strengthens the quality/cost tradeoff: better predictors exist, but they no longer preserve the current systems wedge.

Rank	Cost vs exact sink QK	Abs loss delta	Loss improve	KL improve	Top-1 diff
1	0.266x	0.137	0.031	0.029	0.143
2	0.531x	0.096	0.072	0.069	0.125
4	1.062x	0.062	0.107	0.097	0.095
8	2.125x	0.044	0.125	0.114	0.080

Table 5: Rank frontier on 48 traces, length 96, four sinks, distilgpt2/OPT-125M, seeds 0/1/2. Higher rank reduces downstream drift, but rank 4 and rank 8 exceed exact four-sink QK multiply-add cost, so rank 2 remains the only current systems compromise.

3 Not Claimed

SinkAware does not yet claim benchmark accuracy, cross-model predictor transfer, native GPU speed, HBM savings, latency, throughput, or broad novelty for learned attention sinks. All reported predictors are fit separately per model/config/split. The current contribution is a bounded falsification and quality-control packet: exact static reuse is killed, all-head rank-2 remains alive under Mac-local attention/output/downstream controls, and native timing is now the decisive systems gate.

The novelty boundary is intentionally narrow. Existing systems already expose learned or synthetic sink terms in the softmax denominator, sparse block selection, and retained initial-token caches. SinkAware is not a claim that attention sinks are new; it asks whether fixed early-position K/V sink logits can be approximated cheaply enough to justify a native operator while keeping ordinary non-sink attention exact.

4 Reference and Kernel Gates

The Phase 3 reference specifies the exact computation to preserve during kernel work. Given query q, keys K, values V, and predicted fixed-sink logits $\hat{s}$, SinkAware computes all tail logits $qK_{tail}^{\top}$ exactly, replaces only the first sink-token logits with $\hat{s}$, and then runs the ordinary softmax denominator over the concatenated vector. Unit tests verify that when $\hat{s}$ equals exact sink logits, the reference reproduces exact attention to numerical tolerance. This keeps the live branch separate from the killed exact-static-reuse branch.

The Phase 4 Triton-interpreter scaffold now targets the same operator for a one-head scalar-output synthetic gate. Triton's official debugging documentation describes interpreter mode as a CPU simulation path enabled by `TRITON_INTERPRET=1`, bypassing compilation for debugging.[1] The local Mac environment now uses a repo-local source build of the experimental `triton-cpu` backend, and the SinkAware Phase 4 interpreter tests pass. This closes the Mac kernel-logic correctness gate before native timing work, but it is not evidence of GPU performance.

The native GPU handoff now has an admissibility checker. Returned native packets must include CUDA/NVIDIA metadata, measured model and sequence-shape metadata, quality drift, per-head drift, same-shape latency repeats, NCU memory summaries, and a decision file over the same row/model/shape groups; the checker rejects missing rows, placeholder packets, non-NVIDIA metadata, nonnumeric metrics, incomplete row coverage within any measured model/sequence-length/batch-size group, and fewer than three distinct latency repeats for any row/model/sequence-length/batch-size group, mismatched quality/latency/NCU shapes, or stale model/shape metadata. This is still not a success criterion. It only prevents incomplete native packets from becoming paper evidence.

[1] `https://triton-lang.org/main/programming-guide/chapter-3/debugging.html`

5 Decision

SinkAware is alive only as an all-head approximate low-rank correctness and native-GPU gate. Exact static reuse remains dead, and the simple head-selective rescue is weakened. The larger trace-level frozen repeat strengthens the all-head row, the repeated OPT-family length gate avoids an immediate model-family or length collapse, the Triton interpreter gate passes, and the 48-trace downstream control repeats favor rank-2 over position-only. Promotion now requires a native packet that passes `check_native_gpu_packet.py` and then shows real speed or memory-traffic advantage over exact attention while preserving the downstream-control behavior reported here.

6 Artifacts

- `experimental/sinkaware/phase2/real_qk_sink_softmax_output_probe.md`
- `experimental/sinkaware/phase2/head_selective_sink_gate.md`
- `experimental/sinkaware/phase2/rank2_split_stability_gate.md`
- `experimental/sinkaware/phase2/rank2_length_sink_sweep_gate.md`
- `experimental/sinkaware/phase2/rank2_trace_frozen_split_gate.md`
- `experimental/sinkaware/phase2/rank2_cross_model_falsification_gate.md`
- `experimental/sinkaware/phase2/rank2_cross_model_length_stability_gate.md`
- `experimental/sinkaware/phase2/qk_sink_cost_model.md`
- `experimental/sinkaware/phase3/approx_sink_attention_reference.md`
- `experimental/sinkaware/phase3/downstream_quality_control_gate.md`
- `experimental/sinkaware/phase3/downstream_quality_control_sweep.md`
- `experimental/sinkaware/phase3/downstream_quality_control_gate_traces48_len64_sink2.md`
- `experimental/sinkaware/phase3/downstream_quality_control_gate_traces48_len96_sink2.md`
- `experimental/sinkaware/phase3/downstream_quality_control_gate_traces48_len64_sink4.md`
- `experimental/sinkaware/phase3/downstream_quality_control_gate_traces48_len96_sink4.md`
- `experimental/sinkaware/phase3/downstream_rank_frontier_traces24_len96_sink4.md`
- `experimental/sinkaware/phase3/downstream_rank_frontier_traces48_len96_sink4.md`
- `experimental/sinkaware/phase4/approx_sink_attention_triton_gate.md`
- `experimental/sinkaware/phase2/gpu_gate_runbook.md`
- `experimental/sinkaware/phase2/check_native_gpu_packet.py`
- `experimental/triton_cpu_source_install_20260506.md`
- `experimental/sinkaware/paper/reviewer_pack.md`